

Rotation Solutions

Question 1 (Conceptual)

The rotational kinetic energy (K_{rot}) is given by the equation: $K_{rot} = \frac{1}{2}I\omega^2$ where I is the moment of inertia and ω is the angular speed. Since both discs have the same angular speed (ω), the one with the greater moment of inertia (I) will have the greater rotational kinetic energy.

For a uniform disc, the moment of inertia about its center is:

$$I = \frac{1}{2}MR^2$$

Both discs have identical mass (M), but $R_Y = 1.0 \text{ m}$ and $R_X = 0.5 \text{ m}$. The moment of inertia is proportional to the square of the radius (R^2). Since R_Y is twice R_X , I_Y is four times I_X :

$$I_Y \propto (1.0 \text{ m})^2 = 1.0$$

$$I_X \propto (0.5 \text{ m})^2 = 0.25$$

Since $I_Y > I_X$, **Disc Y** has the greater rotational kinetic energy. This shows that mass distributed farther from the axis of rotation significantly increases the moment of inertia, and thus the rotational energy.

Question 2 (Calculation)

Given values:

- Mass of the sphere (M): 3.0 kg

- Radius of the sphere (R): 0.2 m
- Incline length (L): 5.0 m
- Incline angle (θ): 30°

(a) Write the relationship between translational and rotational kinetic energy for rolling without slipping.

For rolling without slipping, the total kinetic energy (K_{total}) is the sum of the translational kinetic energy (K_{trans}) and the rotational kinetic energy (K_{rot}):

$$K_{total} = K_{trans} + K_{rot}$$

$$K_{total} = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$$

And for rolling without slipping, the relationship between linear speed (v) and angular speed (ω) is: $\omega = \frac{v}{R}$

(b) Calculate the sphere's speed at the bottom.

Use the conservation of energy principle, where the initial potential energy (U_g) is converted to total kinetic energy (K_{total}). $U_{g,i} = K_{total,f}$

First, find the height (h) the sphere drops:

$$h = L \sin(\theta) = (5.0 \text{ m}) \sin(30^\circ) = (5.0 \text{ m})(0.5)$$

$$h = 2.5 \text{ m}$$

The moment of inertia for a solid sphere is $I = \frac{2}{5}MR^2$.

Substitute I and ω into the energy conservation equation:

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}\left(\frac{2}{5}MR^2\right)\left(\frac{v}{R}\right)^2$$

Simplify the rotational term:

$$\frac{1}{2} \left(\frac{2}{5} M R^2 \right) \left(\frac{v^2}{R^2} \right) = \frac{1}{5} M v^2$$

The conservation equation becomes (the mass M cancels out):

$$Mgh = \frac{1}{2} M v^2 + \frac{1}{5} M v^2$$

$$gh = v^2 \left(\frac{1}{2} + \frac{1}{5} \right)$$

$$gh = v^2 \left(\frac{5 + 2}{10} \right)$$

$$gh = \frac{7}{10} v^2$$

Solve for the final speed (v):

$$v = \sqrt{\frac{10}{7} gh}$$

Substitute the values $g = 9.8 \text{ m/s}^2$ and $h = 2.5 \text{ m}$:

$$v = \sqrt{\frac{10}{7} (9.8 \text{ m/s}^2)(2.5 \text{ m})}$$

$$v = \sqrt{\frac{10}{7} (24.5 \text{ m}^2/\text{s}^2)}$$

The speed at the bottom is: $v \approx 5.92 \text{ m/s}$

(c) Determine the total rotational kinetic energy at the bottom.

First, calculate the moment of inertia (I):

$$I = \frac{2}{5} M R^2 = \frac{2}{5} (3.0 \text{ kg})(0.2 \text{ m})^2$$

$$I = \frac{2}{5}(3.0 \text{ kg})(0.04 \text{ m}^2)$$

$$I = \frac{0.24}{5} \text{ kg} \cdot \text{m}^2$$

$$I = 0.048 \text{ kg} \cdot \text{m}^2$$

Next, calculate the angular speed (ω):

$$\omega = \frac{v}{R} = \frac{5.916 \text{ m/s}}{0.2 \text{ m}}$$